Object Knowledge Distillation for Joint Detection and Tracking in Satellite Videos

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***Abstract*— Existing mainstream multiobject tracking (MOT) methods can be categorized into two frameworks, including two- and one-stage ones. Two-stage ones divide MOT task into object detection and association tasks, which usually achieve high accuracy. One-stage ones train a joint model to achieve both detection and tracking. Therefore, their advantage usually lies in the high tracking efficiency. In this article, we inherit the advantages of the two types of frameworks and propose the object knowledge distilled joint detection and tracking framework (OKD-JDT) to achieve accurate as well as efficient tracking. First, the performance of two-stage methods largely depends on the highly performed detection network. Therefore, we treat the detection network as the teacher network to guide the discriminative object feature learning in one-stage methods by using knowledge distillation (KD). Then, in distillation learning, we design adaptive attention learning to learn the discriminative features from the teacher network to student network. In addition, with the similar appearance and uniform moving behavior of objects in satellite videos, we propose to use a joint center point distance and intersection over onion (IOU) to generate tracklets. Experiments on JiLin-1 satellite videos with different objects demonstrate the effectiveness and the state-of-the-art performance of the proposed method.**

***Index Terms*— Knowledge distillation (KD), multiobject tracking (MOT), satellite video.**

I. Introduction

REMOTE sensing image processing plays an important role in Earth observation systems [1], [2], [3], [4]. As a new type of remote sensing image data, the research of satellite videos is of great significance in the fields of military, commercial, engineering, and so on [5], [6], [7]. Nowadays, research on satellite videos includes single-object tracking (SOT) [8], [9], multiobject tracking (MOT) [6], [10], moving object detection (MOD) [7], and so on. In this article, we aim to tackle MOT task in satellite videos, which has a wide range of applications, such as commercial vehicle surveillance, military strikes, and engineering monitoring [6], [10]. The MOT task aims to find the object locations and the trajectory identities across frames in a video [11]. Over the past few decades, MOT tasks have attracted more and more attention and many special methods have been proposed. Multiple hypothesis tracking (MHT) [12] and joint probabilistic data association filter (JPDAF) [13] are two classic traditional methods. Popular methods can be divided into two categories: detection-based tracking (DBT), i.e., the two-stage methods [14], [15], [16], [17], [18], [19], and joint detection and tracking (JDT), i.e., the one-stage methods [20], [21], [22], [23], [24], [25].

The tracking process of the DBT methods usually can be divided into two steps [Fig. 1(a)]: 1) high performed object detectors, such as Faster RCNN [26], YOLO series [27], [28], [29], [30], [31], SDP [32], and DPM [33], are used to obtain the detection results of objects in each frame and 2) then, a separate data association method is used to build the identities of objects across frames. The data association method usually contains similarity measurement and key point matching. The popular similarity measurement methods include appearance feature-based methods [10], [25], intersection-over-onion (IOU)-based methods [19], motion affinity-based methods [34], and so on. The most widely used matching algorithm is Hungarian algorithm [35]. Since each step is independent, the appropriate method can be designed specifically for each stage. In this way, DBT methods usually have high object localization and tracking accuracy. Meanwhile, due to multiple networks in DBT methods, the tracking efficiency is relatively low.

In order to improve the tracking efficiency, JDT methods [23] are proposed and have become one of the mainstream methods. They complete MOT task by simultaneously [23], [36] achieving object detection and identity embedding/re-identification (ReID) in a single network [Fig. 1(b)]. During learning, these methods usually simultaneously optimize objectives of anchor-based detectors such as YOLOv3 [29] or Mask R-CNN [37] and the ReID loss to complete both detection and identity association across frames. CenterTrack [22] was proposed as a point-based framework for JDT. It is based on the CenterNet detector [38] and accomplishes the MOT task by simultaneously optimizing center point-based object detection loss and object offset-based loss between

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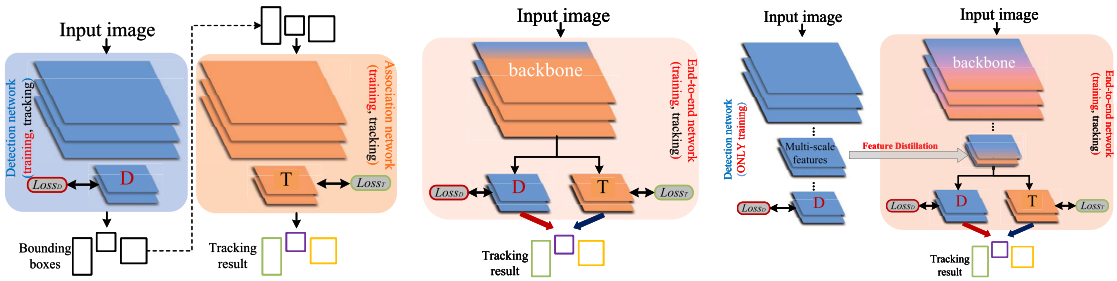


Fig. 1. Tracking flowcharts of DBT-based, JDT-based, and the proposed OKD-JDT frameworks. (a) DBT framework uses separate detection and association networks and it achieves high tracking accuracy due to the highly performed detection network. (b) JDT framework shares one backbone for detection and association, and thus, the tracking efficiency is much higher. (c) Proposed framework uses the JDT for tracking while distilling object features from the highly performed detection network, which inherits advantages of both DBT and JDT frameworks.

two adjacent frames. However, the above methods usually operate in the current frame and the detector may have missed detections in subsequent frames. Then, TraDeS [25] was proposed, which estimates the offset distance of the tracked object based on the cost volume. It extracts the features of the objects through forward propagation to improve the object detection performance in the current frame via objects in the last frame. TGraM [10] improved TraDeS by the graph-based structure. Since such methods accomplish multitask learning by sharing a single network, such methods usually have high tracking speed. However, point-based object detection often results in lower tracking accuracy than DBT ones.

As analyzed above, DBT methods usually have high tracking accuracy, while JDT methods usually have high tracking speed. In this article, in order to inherit the advantages of both the two frameworks, we propose to use a specialized object detection network as the teacher network to guide the object feature learning of the multitask learning in a JDT network [Fig. 1(c)], and thus, we name the method as the object knowledge distilled JDT method, OKD-JDT for short. In addition, considering that the spatial resolution of satellite videos is much lower than natural videos, which results in a similar appearance between objects and the top-down view of satellites, we propose to adopt joint center point distance and IOU to generate tracklets. On the one hand, due to the knowledge distillation (KD) of the specialized object detection network, the object locating capability of the joint network will be improved. On the other hand, the multitask learning adopted can lead to a high tracking speed by sharing the joint network.

In summary, the main contributions of this article can be summarized as the following three aspects.

1) To improve tracking accuracy and meanwhile maintain tracking efficiency, an object knowledge distilled JDT framework (OKD-JDT) is proposed by distilling the object discriminative features from a highly performed object detection network to a one-stage method. Since the distillation is only implemented in training, the tracking efficiency is not influenced.
2) In order to better distilling the appropriate features from the teacher network to the student network, we design adaptive attention learning to learn the distillation weights.
3) We propose to adopt joint center point distance and IOU to generate tracklets to deal with similar appearances between different targets.

The rest of this article is organized as follows. Section II introduces the related work. Section III introduces the proposed method. Section IV presents the experimental setup and experimental results in detail. Finally, Section V summarizes this article.

II. Related Work

A. MOT in Satellite Videos

With the rapid development of remote sensing technology, remote sensing video has appeared in recent years. Research on remote sensing video has broad application prospects in military, commercial, and other fields. MOT methods for remote sensing videos have also been extensively researched [6], [39], [40], [41], [42], [43]. To guarantee the tracking accuracy, most of them are DBT-based ones by considering the properties of satellite videos. To reduce false alarms, missed objects, ID switches, and low-confidence bounding boxes, SFMFMOT [42] was proposed by considering both slow and motion features. To improve the performance of crowded similar vehicle detection, in [40], a customized cross-frame-based object detection network and a two-branch long short-term memory network were proposed to exploit complementary information between frames and motion information, respectively. A dataset named VISO for MOT and MOD was constructed in [6] where a motion modeling baseline was also proposed via accumulative multiframe differencing and matrix completion to improve the performance. A novel detecting algorithm by an exponential probability distribution based on the local noise modeling was proposed in [43]. Also, a multi-morphological cue in [43] is designed to further improve the performance. To reduce detector degradation, a bidirectional MOT framework was proposed in [39] based on SOT with trajectory criteria (BMTC). Since tracking efficiency is also critical for satellite computing devices, JDT-based methods have also been developed. TGraM [41], a JDT method, considers tracking as a graph information reasoning procedure and deals it with multitask learning. In this article, we propose a new tracking framework that inherits the advantages of both DBT and JDT frameworks. The key idea is to use the knowledge of a highly performed detection network to increase the detection performance of an end-to-end tracking network.

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B. KD and Its Applications

In recent years, due to its powerful scalability and the ability to encode large-scale data, deep learning has been a huge success in the field of academia and industry [44], [45], [46], [47]. However, due to resource capacity, deep models are difficult to efficiently implement in practical applications, such as on embedded and mobile devices [48]. To solve this difficulty, a large number of model compression and acceleration technologies [49], [50] have emerged, and KD [51] is one of them, which can effectively learn from large teacher models to small student models. The main idea of KD is that the student model imitates the teacher model [51] or the two networks compete with each other [52] so that the student model can perform equally to or even better than the teacher model. Recently, many methods based on KD have been widely used in the computer vision field, such as semantic segmentation [53], face recognition [54], behavior prediction [55], SOT [56], and MOT [57].

Different from the existing multitarget tracking method based on KD in [57], which distills ReID to improve the association performance, the proposed method distills the object detection information to improve the location accuracy of the JDT framework. Specifically, in this article, in order to maintain high tracking speed and meanwhile achieve high tracking accuracy, considering the benefit of KD, we propose to use feature-based offline distillation to learn the high-precision object locating in satellite videos based on a one-stage MOT framework. Considering that single-stage networks can improve tracking efficiency by sharing parameters, our method is expected to achieve fast and accurate tracking with the benefit of object feature distillation.

III. Proposed Method

The proposed method adopts the classic CenterTrack [22] as the baseline. Therefore, in this section, we first review some details of CenterTrack.

A. Revisit of CenterTrack

CenterTrack [22] is a point-based JDT framework. During tracking, for the tth frame, it detects objects $T^{(t)} = \{b_0^{(t)}, b_1^{(t)}, \dots, \}$ and assigns the attributes for each object $b = (\mathbf{p}, \mathbf{s}, w, id)$, which is described by its center location $\mathbf{p} \in \mathbb{R}^2$, size $\mathbf{s} \in \mathbb{R}^2$, detection confidence $w \in [0, 1]$, and a unique identity $id \in \mathbb{I}$ given the tth frame $I^{(t)} \in \mathbb{R}^{H \times W \times 3}$, previous frame $I^{(t-1)} \in \mathbb{R}^{H \times W \times 3}$, and tracked objects in the previous frame $T^{(t-1)} = \{b_0^{(t-1)}, b_1^{(t-1)}, \dots, \}$.

The backbone network of CenterTrack is first trained via a large-scale tracking dataset. With the locations of annotated objects $\{\mathbf{p}_0, \mathbf{p}_1, \dots, \}$ in an image, CenterTrack uses the focal loss [58] formulated in (1) as the objective and produces a low-resolution heatmap $\hat{Y} \in [0, 1]^{(W/R) \times (H/R) \times C}$ $(R = 4)$

$$L_k = \frac{1}{N} \sum_{xyz} \begin{cases} (1 - \hat{Y}_{xyz})^{\alpha} \log(\hat{Y}_{xyz}), & \text{if } Y_{xyz} = 1 \\ (1 - Y_{xyz})^{\beta} (\hat{Y}_{xyz})^{\alpha} \log(1 - \hat{Y}_{xyz}), & \text{otherwise} \end{cases} \tag{1}$$

where $Y \in [0, 1]^{(W/R) \times (H/R) \times C}$ is a ground truth (GT) with the annotated objects. x, y, and z denote the indexes in the dimensions of width, height, and channel, respectively. N is the number of all objects. $\alpha = 2$ and $\beta = 2$ are user-defined hyperparameters of the focal loss. In point-based methods, each detected object is taken as a single center point, and therefore, CenterTrack renders all detections in a class-agnostic single-channel heatmap using the Gaussian render function. CenterTrack renders each point $\mathbf{p}$ of class c into a Gaussian-shaped peak $Y_{:,:,c}$ using a rendering function $Y = R_{\mathbf{q}}(\mathbf{p}_0, \mathbf{p}_1, \dots,)$ defined as

$$R_{\mathbf{q}}(\mathbf{p}_0, \mathbf{p}_1, \dots,) = \max_i \exp\left(-\frac{(\mathbf{p}_i - \mathbf{q})^2}{2\sigma_i^2}\right) \tag{2}$$

where $\mathbf{q} \in \mathbb{R}^2$ denotes the position index in the Gaussian-shaped map. The Gaussian kernel σ_i is a function of the object size.

CenterTrack also produces a size prediction $\hat{S} \in \mathbb{R}^{(W/R) \times (H/R) \times 2}$, which is determined by the center locations. With the bounding box size $\mathbf{s}_i$ of the ith object at location $\mathbf{p}_i$, the size prediction is learned by the following regression problem:

$$L_{\text{size}} = \frac{1}{N} \sum_{i=1}^{N} \left| \hat{S}_{\mathbf{p}_i} - \mathbf{s}_i \right|. \tag{3}$$

In addition to object localization, CenterTrack simultaneously learns the association of detected objects. It introduces two additional output channels $\hat{D}^{(t)} \in \mathbb{R}^{(W/R) \times (H/R) \times C}$ as the predicted 2-D displacement. Then, the displacement $\hat{\mathbf{d}}^{(t)} = \hat{D}^{(t)}_{\hat{\mathbf{p}}^{(t)}}$ for each detected object at location $\hat{\mathbf{p}}^{(t)}$ generates the difference in location of the object in the current frame $\hat{\mathbf{p}}^{(t)}$ and the previous frame $\hat{\mathbf{p}}^{(t-1)}$: $\hat{\mathbf{d}}^{(t)} = \hat{\mathbf{d}}^{(t)} - \hat{\mathbf{d}}^{(t-1)}$. This displacement is trained by minimizing the following regression problem:

$$L_{\text{off}} = \frac{1}{N} \sum_{i=1}^{N} \left| \hat{D}_{\mathbf{p}_i^{(t)}} - \left(\mathbf{p}_i^{(t-1)} - \mathbf{p}_i^{(t)} \right) \right| \tag{4}$$

where $\mathbf{p}_i^{(t-1)}$ and $\mathbf{p}_i^{(t)}$ are GT objects.

Moreover, to improve the localization accuracy, CenterTrack introduces a refined center local location for regression using an analogous L_1 loss

$$L_{\text{loc}} = \frac{1}{N} \sum_{\mathbf{p}} \left| \hat{O}_{\tilde{\mathbf{p}}} - \left(\frac{\mathbf{p}}{R} - \tilde{\mathbf{p}} \right) \right| \tag{5}$$

where $\hat{O}_{\tilde{\mathbf{p}}}$ represents a local offset for each center point. This loss function is used to reduce the discretization error caused by the output stride. The loss function is used only to keypoint locations $\tilde{\mathbf{p}}$ which is the floor int of $\lfloor (\mathbf{p}/R) \rfloor$, i.e., $\tilde{\mathbf{p}} = \lfloor (\mathbf{p}/R) \rfloor$, while all other locations are ignored.

The overall training objective is:

$$L_{\text{CTracker}} = L_k + \lambda_{\text{size}} L_{\text{size}} + \lambda_{\text{off}} L_{\text{off}} + \lambda_{\text{loc}} L_{\text{loc}} \tag{6}$$

where λ_{size}, λ_{off}, and λ_{loc} represent weight control parameters.

During tracking, after detecting all the objects, in descending order of confidence $\hat{w}$, CenterTrack greedily associates each position $\hat{\mathbf{p}}$ with the closest unmatched prior detection at

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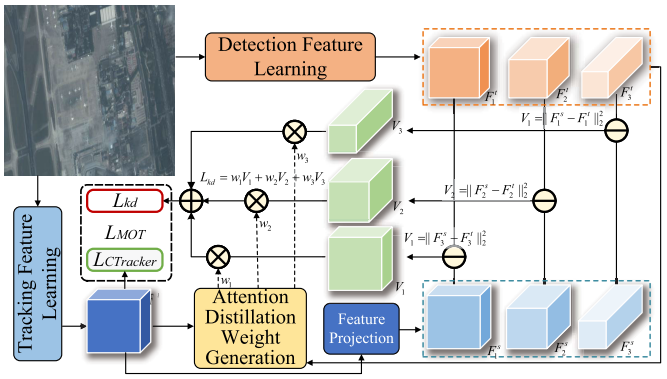


Fig. 2. Overview of the proposed method.

position $\hat{\mathbf{p}} - \hat{D}_{\hat{\mathbf{p}}}$. A new tracklet will be generated if there is no unmatched prior detection within a radius defined as κ.

B. Overall Architecture

CenterTrack uses multitask learning for detection and data-related tasks. Through the sharing of backbone network parameters, it has high tracking efficiency. However, due to point-based detection, the detection performance is easily disturbed and the detection performance is limited. In order to improve the detection performance and then improve the tracking performance, we propose to use a special object detection network (teacher network, namely, YOLOX [31]) to guide the feature learning of CenterTrack to improve the accuracy of locating and then improve the tracking performance.

As shown in Fig. 2, the proposed method takes an image $I \in \mathbb{R}^{H\times W\times 3}$ as input. The feature map F_l^t from the teacher network is obtained through the YOLOX branch. The knowledge from such features is expected to be distilled to the features of a tracking network (student network), and here, we use the CenterTrack as our baseline. Then, the feature map F^s from the student network is also obtained through the CenterTrack backbone network

$$F^s = \text{CenterTrack}_{\text{backbone_network}}(I) \tag{7}$$

$$F_1^t, F_2^t, \ldots, F_l^t = \text{YOLOX}_{\text{backbone_network}}(I). \tag{8}$$

Then, the features of F^s are enhanced by distilling the features of $F_1^t, F_2^t, \ldots, F_l^t$ into F^s. We propose to use adaptive attention-weight learning to achieve feature distillation. As for the objective function, in addition to optimizing L_{CTracker}, the proposed method also optimizes the proposed KD loss, namely, L_{kd}. Therefore, in the following, attention distillation weight generation, objective function, optimization, and so on are introduced in detail.

C. Adaptive Attention Weight Generation

Since the teacher and student networks are focusing on different tasks, it is necessary to distill the critical features that can facilitate the detection performance while not significantly influencing the tracking process. Here, we try to learn weights to capture the critical features. As a consequence, we propose an adaptive attention learning method to generate the adaptive weights that guide critical features from teacher to student networks (Fig. 3).

In order to improve efficiency and maintain good performance, we select the last three layers (F_1^t, F_2^t, and F_3^t) of feature maps in the teacher network as the distillation source feature map. Given the feature maps F^s, F_1^t, F_2^t, and F_3^t, to obtain adaptive weights, we first compute the point-to-point similarity between each pair of components in the feature maps and obtain the similarity matrix M^s, M_1^t, M_2^t, and M_3^t through pairwise similarity calculation. From the feature map to the similarity matrix, we first reshape the feature maps into a row vector $reshape(F)$. Then, the inner product is computed

$$M^s = \text{reshape}\left(F^s\right) \odot \text{reshape}\left(F^s\right)^T \tag{9}$$

$$M_1^t = \text{reshape}\left(F_1^t\right) \odot \text{reshape}\left(F_1^t\right)^T \tag{10}$$

$$M_2^t = \text{reshape}\left(F_2^t\right) \odot \text{reshape}\left(F_2^t\right)^T \tag{11}$$

$$M_3^t = \text{reshape}\left(F_3^t\right) \odot \text{reshape}\left(F_3^t\right)^T \tag{12}$$

where ^{T} represents the transpose operation. However, in order to obtain a unified dimension to calculate the similarity weight between the feature maps of the student and the teacher networks, the similarity matrix is needed to input into the multilayer perceptron (MLP) structure to output vectors of the same dimension. Note that the MLP shares parameters for each similarity matrix. Through MLP, the output query (Q^s) and key (K_1^t, K_2^t, K_3^t) vectors are obtained

$$Q^s = \text{MLP}\left(M^s\right) \tag{13}$$

$$K_1^t = \text{MLP}\left(M_1^t\right) \tag{14}$$

$$K_2^t = \text{MLP}\left(M_2^t\right) \tag{15}$$

$$K_3^t = \text{MLP}\left(M_3^t\right). \tag{16}$$

After obtaining the key and query vectors, the widely used Softmax function is used to calculate the similarity weight, specifically

$$w_1\left(F^s, F_1^t\right) = \frac{e^{(Q^s)^T K_1^t}}{e^{(Q^s)^T K_1^t} + e^{(Q^s)^T K_2^t} + e^{(Q^s)^T K_3^t}} \tag{17}$$

$$w_2\left(F^s, F_2^t\right) = \frac{e^{(Q^s)^T K_2^t}}{e^{(Q^s)^T K_1^t} + e^{(Q^s)^T K_2^t} + e^{(Q^s)^T K_3^t}} \tag{18}$$

$$w_3\left(F^s, F_3^t\right) = \frac{e^{(Q^s)^T K_3^t}}{e^{(Q^s)^T K_1^t} + e^{(Q^s)^T K_2^t} + e^{(Q^s)^T K_3^t}}. \tag{19}$$

After obtaining the attention weights, next, the objective function and optimization of the proposed method are introduced.

D. Adaptive Distillation Learning

1) Objective Function: Since we aim to use KD to distill feature maps in the object detection network to guide the learning of features for the student tracking network, the distillation loss L_{kd} is designed as the weighted distance between feature maps of teacher network and projected features from F^s.

Given the feature maps F^s and F_1^t, and F_2^t and F_3^t, in order to facilitate the loss calculation, we first project the feature map F^s into F_1^s, F_2^s, and F_3^s with the same dimension to F_1^t, F_2^t, and F_3^t

$$\left\{F_1^s, F_2^s, F_3^s\right\} = \text{projection_function}\left(F^s\right) \tag{20}$$

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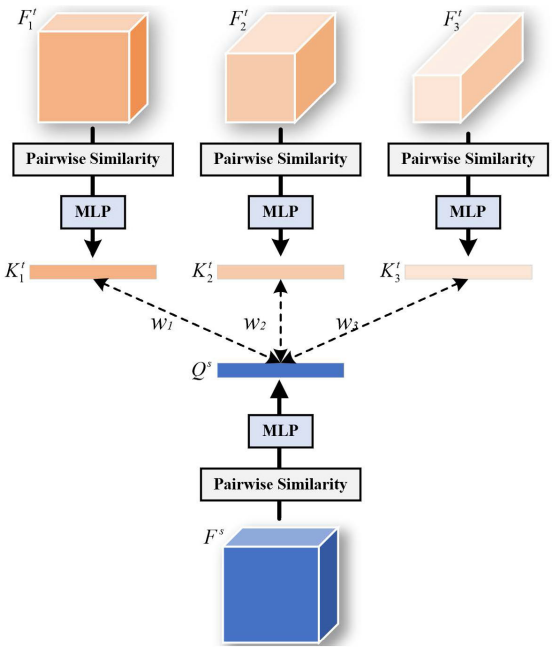


Fig. 3. Overview of the KD module.

where projection_function is implemented by a stack of three convolutional layers with $\text{conv}_{1\times1}$, $\text{conv}_{3\times3}$, and $\text{conv}_{1\times1}$.

As discussed above, here, we use the mean square error (Euclidean distance) to compute the loss between feature maps of teacher and student networks in different scales

$$V_1 = \|F_1^s - F_1^t\|_2^2 \tag{21}$$

$$V_2 = \|F_2^s - F_2^t\|_2^2 \tag{22}$$

$$V_3 = \|F_3^s - F_3^t\|_2^2 \tag{23}$$

where V_1–V_3 denote the distance in different feature scales. Then, to distill critical features, the similarity weights w_1–w_3 computed in (17)–(19), respectively, are multiplied with V to get the distillation loss function L_{kd}

$$L_{\text{kd}} = w_1 V_1 + w_2 V_2 + w_3 V_3. \tag{24}$$

The overall MOT objective function of the proposed method is generated

$$L_{\text{MOT}} = L_{\text{CTracker}} + \lambda_{\text{kd}} L_{\text{kd}} \tag{25}$$

where λ_{kd} controls the importance of distillation loss.

2) Optimization: The optimization follows a backpropagation algorithm where the three networks, including detection feature learning network, tracking feature network, and the attention distillation weight generation network, should be optimized. As the teacher network, the feature learning network is first trained following the YOLOX, and the feature maps F_1^t, F_2^t, and F_3^t are generated for object distillation. With the gradient of L_{kd}, i.e., δL_{kd}, the gradient of w_1–w_3, which are outputs of the attention distillation weight generation network, can be obtained: $\delta w_1 = \delta L_{\text{kd}} V_1$ as well as δw_2 and δw_3. Through the backpropagation of the attention distillation weight generation network, we can obtain the gradient of the input F^s, i.e., δAF^s. Moreover, the distillation operator also generates backpropagated gradients δDF_i^s with $i = 1, 2, 3$

$$\delta DF_i^s = \delta L_{\text{kd}} w_i \left(F_i^s - F_i^t\right). \tag{26}$$

For the tracking feature learning network, the output gradient δF^s is computed via three gradients, including $\delta L_{\text{CTracker}}$, δAF^s, and the δDF^s obtained by backprojection from $\delta DF_i^s, i = 1, 2, 3$: $\delta F^s = \delta L_{\text{CTracker}} + \lambda_{\text{kd}}(\delta AF^s + \delta DF^s)$. Then, all the networks can be updated via the gradients.

E. Tracklet Generation

Similar to one-stage methods in [41] and [25], for a prediction box in frame t, first, we get the prediction center position of the previous frame according to the prediction offset. Then, the distance of each prediction center between the current and last frames within the trajectory box is computed as the matching cost matrix. Meanwhile, the size of the corresponding prediction box and the trajectory box is used to limit the matching range. Specifically, the cost matrix value is set to infinity for those distances greater than any size of the prediction box and the trajectory box. Finally, we use the greedy algorithm to associate the trajectory box and the prediction box according to the cost matrix.

However, the above process uses a single offset prediction head to predict the offsets of all detections. However, there may be biases between different objects, especially when the objects are moving at different speeds. To solve the problem of offset prediction bias, which potentially leads to the failure of the first round of matching, we design a simple IOU-based secondary matching strategy. Specifically, we use the IOU of the prediction in the current frame and trajectory box in the last frame as the cost matrix and the Hungarian algorithm is used to make a match. In addition, in order to enlarge the matching range, we set the matching threshold of IOU to be 0.5. It can be seen from the experimental results in Section IV that the matching effectively alleviates the problem of long-distance mismatching.

Finally, we update the trajectory boxes with the prediction boxes on the match. For the detected target that fails to match after both rounds of matching, if the confidence of the detection is greater than a predefined threshold δ, it is added as a new trajectory. For unmatched trajectory, we can directly use the predicted offset to supplement the miss detection. However, it may also result in errors when the object is lost in real cases. Therefore, we directly delete the unmatched trajectory. Although this may result in frequent ID switches, it also reduces false detections as shown in experiments (Section IV).

F. Effectiveness Analysis

In Fig. 4, we visualize several feature maps of the same frame image before and after adding the KD module. The extracted feature maps enable a clearer distinction between the environment background and the target without compromising background integrity and specific textures. It can also be seen from Fig. 4 that when the effective features of the extracted target are clearer, the extraction of interference information in the environment will be reduced, which potentially reduce the false positives (FPs) and false negatives (FNs). The ablation experiment in Section IV-D also indicated that the number of FPs and FNs can be significantly reduced by adding KD

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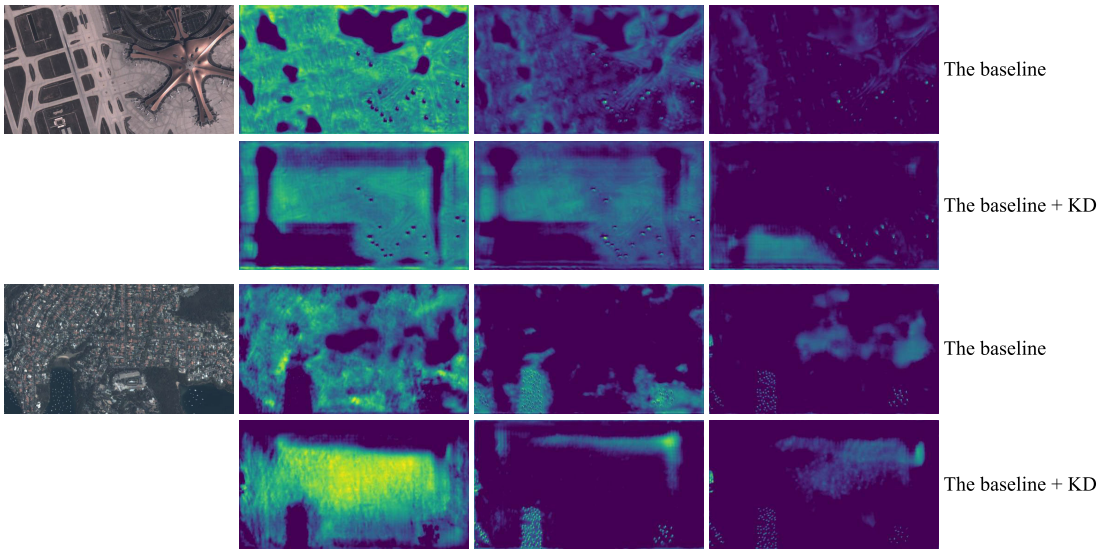


Fig. 4. Feature maps visualization of the baseline and the baseline + KD on two video frames.

modules, thereby improving the detection (IDP and IDR) and tracking accuracy (MOTA) of the proposed method.

IV. Experimental Study

In order to verify the performance of the proposed method, we construct experiments on remote sensing videos in different regions and with different objects. In this section, we first introduce the dataset used, evaluation metrics, and implementation details. Then, effectiveness validation and hyperparameter experiments are described to demonstrate the superiority over baseline. Finally, the state-of-the-art comparison is conducted to show the state-of-the-art performance of the proposed method.

A. Dataset Description

To well verify the performance of the proposed method, we conduct experiments on three publicly available datasets, namely, AIR-MOT [41], VISO [6], and SAT-MTB [59], which are described as follows.

The videos in AIR-MOT [41] are taken from the satellite Jilin-1 from October 2017 to October 2020 in Abu Dhabi, Beijing, Dubai, Los Angeles, Sanya, San Diego, Shanghai, Sydney, and Yokohama. The spatial resolution is from 0.91 to 1.27 m. The AIR-MOT dataset contains 59 videos that have targets. The frame rate is about 5–10 frames per second (FPS). The frame size is 1920 × 1080 for all videos. The AIR-MOT contains both aircraft and ship targets. The aircraft are relatively large and sparsely arranged. The ships are relatively small and densely arranged and meanwhile arranged in a complex environment. In order to better evaluate the performance of the proposed method in different scenarios, we divide AIR-MOT into two subsets, namely, AIR-ship and AIR-aircraft. In this way, the strengths and weaknesses of the approaches can be more clearly verified.

The videos in VISO [6] are taken from the satellite Jilin-1. The spatial resolution is from 0.91 m. The VISO dataset contains 47 videos that have targets. The frame rate is about 10 FPS. The frame size is 1000 × 1000 for all videos.

The videos in SAT-MTB [59] are taken from the satellite Jilin-1, including China, USA, Mexico, France, Italy, Spain, Australia, Japan, Turkey, India, Thailand, Saudi Arabia, and United Arab Emirates (UAE), from 2017 to 2021. The spatial resolution is 0.92 m. The SAT-MTB dataset contains 249 videos that have targets. The frame rate is about 10 FPS. The frame size is from 512×512 to 1080×2160 for all videos.

Table I shows the details about these three datasets. For a fair comparison, for all experiments, for AIR-aircraft and AIR-ship, the training rate is 50% and the rest is 50% for testing; for SAT-MTB-airplane and SAT-MTB-ship, the training rate is 60% and the rest is 40% for testing; and for VISO-car, the training rate is 30% and the rest is 70% for testing.

B. Evaluation Metrics

As in many MOT tracking papers [22], [41], we also use the py-motmetrics toolbox to evaluate the performance of the proposed and compared methods. The quantitative metrics in py-motmetrics toolbox used in this article, including MOT accuracy (MOTA), identification F-score (IDF1), ID switches (IDs), identification precision (IDP), identification recall (IDR), mostly tracked (MT), mostly lost (ML), FP, FN and fragmentation (FM), among which

$$\text{MOTA} = 1 - \frac{\text{FN} + \text{FP} + \text{IDSW}}{\text{GT}} \in (-\infty, 1] \tag{27}$$

where FP represents false positives, which counts the sum of false positives for the entire video. FN represents false negatives, which counts the sum of false negatives for the entire video. IDSW stands for the sum of the ID switches (IDs) and also counts the number of IDs in the entire video. GT represents the number of GT objects, which counts the sum of GT objects for the entire video. The closer the MOTA is to 1, the better the tracker performance is. Note that due to the number of transitions, the MOTA may be less than 0. IDF1 stands for identification F-score, which refers to the F value identified by the ID in each box

$$\text{IDF1} = \frac{2\text{IDTP}}{2\text{IDTP} + \text{IDFP} + \text{IDFN}} \tag{28}$$

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TABLE I
DATASETS DESCRIPTION

Data sets	Frame image size	Sequence Number	Mean frame number	Mean IDs	Mean object size	Platform	Spatial resolution
AIR-aircraft	1080×1920	31	242	6	35×34	JiLin-1	0.91m - 1.27m
AIR-ship	1080×1920	29	299	20	28×27	JiLin-1	0.91m - 1.27m
SAT-MTB-airpalne	512×512∼1080×2160	62	211	2	50×50	JiLin-1	0.92m
SAT-MTB-ship	512×502∼1080×2160	70	247	4	20×21	JiLin-1	0.92m
VISO-car	1024×1024∼1025×1025	10	317	51	7×7	JiLin-1	0.92m

TABLE II
COMPARISON RESULTS OF THE BASELINE, BASELINE + KD, BASELINE + DA, AND THE PROPOSED METHODS ON DATASET AIR-AIRCRAFT

Methods	MOTA↑	IDF1↑	IDs↓	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	FM↓
Baseline	83.5%	90.2%	282	93.8%	86.9%	176	17	1028	2872	64
Baseline + KD	86.7%	91.3%	275	94.6%	88.2%	179	13	**826**	2520	49
Baseline + DA	84.5%	91.2%	28	94.7%	87.8%	176	17	1028	2872	64
The Proposed-V1	89.2%	94.1%	22	93.9%	94.2%	187	7	1401	1314	40
The Proposed-V2	**91.8%**	**95.7%**	**9**	**96.2%**	**95.2%**	**189**	**5**	897	**1164**	**34**

where IDTP and IDFP represent the number of true IDs and FP IDs, respectively, and IDFN is the number of FN IDs. IDP refers to the accuracy of ID identification in each box. IDR refers to the recall rate identified by the ID in each box. MT represents a trajectory that is tracked to more than 80%. ML stands for a trajectory that is only tracked below 20%. FM indicates the number of times a trajectory has been cut off.

C. Implementation Details

In our experiments, CenterTrack [22] is served as the basic student tracking network and YOLOX [31] served as the teacher network to guide the feature learning of the student network. To accommodate the input and output needs of the teacher–student network and to better match the distillation module, we scale each frame of the original video frame to 1088×608 for AIR-MOT, where 16 are randomly selected as input batch. For VISO, we use the original size of the frames as input size, and for SAT-MTB, we randomly crop frames into 512×512 patches to train the tracking methods. We use the deep layer aggregation (DLA-34) [40] network as the backbone of the student network, which progressively aggregates multiscale features to compute the output features. It can better propagate features and gradients for learning useful features. Moreover, we follow the original training configuration of CenterTrack and train the proposed method with 70 epochs using the Adam optimizer, with the learning rate initially set to $1.25e^{-4}$ and reduced to $1.25e^{-5}$ at 60 epochs. We set the weight parameter λ_{kd} to 0.8 for all compared experiments. In addition, we follow the original configuration setting in CenterTrack, namely, using random rotation, cropping, and color dithering to enhance the image. Specifically, for SAT-MTB, we normalize the stride to 2 if it is larger than 2 to balance the large size variance between objects as in [59]. The experiments are implemented on the platform of Pytorch in the Ubuntu system with the device that includes an RTX 3090 GPU (24-GB memory) and Intel Core i7-11700 CPU.

D. Effectiveness Validation

In order to verify the effectiveness of the proposed method, we compare it with the baseline (CenterTrack), the baseline + KD, the baseline + DA (DA stands for proposed trajectory generation method), and the baseline + KD + DA, namely, the proposed method-$V2$ on the AIR-aircraft and AIR-ship. The proposed-$V1$ represents F_1^S, F_2^S, and F_3^S of DLA-34 backbone in CenterTrack is naturally distilled from corresponding F_1^t, F_2^t, and F_3^t in YOLOX. We use YOLOX as the teacher network and randomly select 10% as the training data to fine-tune the network. Then, the fine-tuned YOLOX model is used as the teacher network to guide the feature learning of the CenterTrack baseline network. Tables II and III show the quantitative comparison results.

From Table II, we can find that in terms of MOTA, the baseline + KD improves the baseline with 3.2%. As for IDF1, the baseline + KD improves the baseline with 1.1%. With KD, the IDs is reduced from 282 to 275. For other indicators such as IDP, IDR, MT, ML, FP, FN, and FM, the baseline + KD generally improves the performance of the baseline. This demonstrates the effectiveness of KD adopted in this article. Then, the baseline + DA also improves the performance of the baseline overall; in particular, the ID is greatly reduced. This further proves the effectiveness of the proposed tracklet generation approach. When the baseline combines KD and DA, the performance is significantly improved. In terms of MOTA, the performance is improved by 8.3%. As for IDFA, 5.5% is improved. Metric ID is reduced from 282 to 9. In addition, most of the other indicators have also been improved.

As shown in Table III, in terms of indicators MOTA, IDF1, IDs, IDP, IDR, MT, ML, FP, FN, and FM, the performance improvement of the baseline + KD over the baseline is: 3.2%↑, 1.0%↑, 89↓, 1.5%↑, 0.7%↑, 4↑, 11↓, 1596↓, 1237↓, and 20↑, respectively. This demonstrates the effectiveness of the proposed method with KD on the AIR-ship dataset. Comparing the baseline with the baseline + DA, we can find that the performance is greatly improved, including the main

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TABLE III
COMPARISON RESULTS OF THE BASELINE, BASELINE + KD, BASELINE + DA, AND THE PROPOSED METHODS ON DATASET AIR-SHIP

Methods	MOTA↑	IDF1↑	IDs↓	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	FM↓
Baseline	27.4%	40.2%	13498	52.3%	32.7%	216	208	7044	36647	**602**
Baseline + KD	30.6%	41.2%	13797	53.8%	33.4%	220	197	5448	35410	622
Baseline + DA	44.4%	60.1%	**109**	78.2%	48.8%	216	208	7044	36647	**602**
The Proposed-V1	59.5%	68.1%	284	79.7%	59.4%	310	138	5808	25822	761
The Proposed-V2	**61.5%**	**70.6%**	194	**83.3%**	**61.3%**	**313**	**134**	**4660**	**25511**	624

TABLE IV
PERFORMANCE COMPARISON OF THE STUDENT MODEL AND THE TEACHER MODEL

Data	Model	TP↑	FP↓	Recall↑	Precision↑	mAP↑	FPS	training time	training memory
AIR-ship	YOLOX (teacher)	**59243**	512107	**75.21%**	10.37%	**66.28%**	55.3	-	-
	CenterTrack	52018	6720	66.04%	88.56%	59.61%	21.8	4.9h	17.2GB
	The proposed	53086	**4583**	67.39%	**92.05%**	63.07%	20.2	5.6h	17.7GB
AIR-aircraft	YOLOX (teacher)	**24237**	233328	**95.70%**	9.41%	90.83%	54.8	-	-
	CenterTrack	23819	1198	94.05%	95.21%	91.27%	21.5	5.2h	17.2GB
	The proposed	24168	**903**	95.43%	**96.40%**	**92.56%**	22.9	5.7h	17.7GB

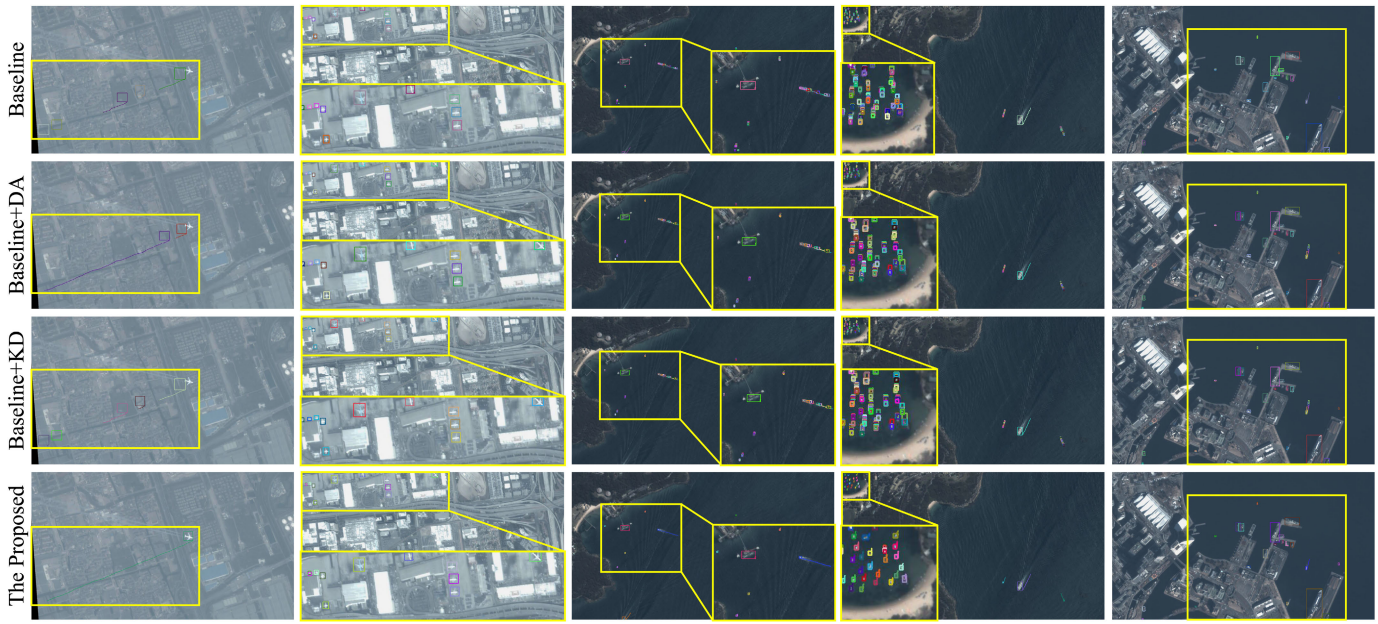


Fig. 5. Tracking results visualization of the baseline, baseline + KD, baseline + DA, and the proposed method on dataset AIR-aircraft.

metrics MOTA, IDF1 or IDs, and all other indicators. In the case of MOTA, the performance improvement is 17%. In terms of IDF1, the performance improvement is 19.9%. IDs is also reduced from 134 498 to 109. Since the ships are usually densely arranged, the proposed trajectory generation method greatly improves the performance. When the baseline fused both KD and DA, MOTA is improved from 27.4% to 61.5%. IDF1 improves from 40.2% to 70.6%. IDs drops from more than 10 000 to below 200. Other indicators have also improved significantly. This further demonstrates the good performance of the proposed method on the AIR-ship dataset.

Finally, we visualize the tracking results of some videos on the baseline, the baseline + KD, the baseline + DA, and the baseline + KD + DA. Fig. 5 shows the comparison results where some parts are enlarged to clearly show the objects and trajectories. From the results, the method that only with KD cannot guarantee the continuity of the detected objects, such as the dense ships and aircraft. By using both KD and DA, the proposed method achieves accurate detection and continuous trajectories, which shows superior performance over the baseline.

In addition, in order to better verify that the teacher network could guide the student network to learn more accurately in the object locating performance, we conduct a comparative experiment between the teacher network and the student network on indicators true positive (TP), FP, recall, precision, mean average precision (mAP), and FPS. Moreover, the training time and memory usage are also listed. The experimental results are presented in Table IV. Note that here, only object localization performance is listed to demonstrate the effectiveness of the proposed KD method. YOLOX is specially designed and trained for object localization. Then, we use it as the teacher network and distill the object features for accurate tracking. With the distilled features, the proposed

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TABLE V
COMPARISON WITH STATE-OF-THE-ART APPROACHES ON DATASET AIR

On **AIR-aircraft**.

Methods	Publication year	DBT/JDT	MOTA↑	IDF1↑	IDs↓	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	FM↓
TraDeS [25]	CVPR2021	JDT	73.6%	84.7%	42	97.4%	75.0%	148	44	418	6233	114
TrackFormer [61]	CVPR2022	JDT	80.2%	88.8%	97	87.1%	90.5%	184	12	2943	1965	110
UMA [20]	CVPR2020	JDT	85.4%	92.1%	19	92.8%	91.4%	186	11	1644	2037	81
FairMOT [24]	IJCV2021	JDT	86.7%	93.2%	23	94.2%	92.2%	180	8	1395	1945	226
StrongSORT [18]	TMM2023	DBT	87.8%	93.5%	**1**	**99.6%**	88.2%	182	18	**86**	2994	**14**
SGT [62]	WACV2023	JDT	89.2%	94.3%	56	94.6%	93.9%	184	6	1252	1439	153
CenterTrack [22]	ECCV2020	JDT	89.3%	94.2%	12	94.8%	93.7%	184	9	1198	1506	35
GHOST [63]	CVPR2023	DBT	89.6%	94.3%	2	98.8%	90.1%	183	15	207	2432	67
SORT [14]	ICIP2016	DBT	89.7%	94.5%	7	99.5%	90.0%	183	17	94	2515	16
CKDNet-SMTNet [40]	ISPRS2021	DBT	82.1%	90.0%	36	85.0%	**95.7%**	187	**3**	3850	**653**	52
DSFNet-SORT [64]	TGRS2023	DBT	82.3%	90.3%	6	98.9%	83.1%	164	28	218	4269	96
TGraM [10]	TGRS2022	JDT	86.3%	92.4%	28	94.1%	90.7%	174	8	1264	2167	181
The Proposed	-	JDT	**91.8%**	**95.6%**	12	96.1%	95.2%	**189**	5	903	1157	34

On **AIR-ship**.

Methods	Publication year	DBT/JDT	MOTA↑	IDF1↑	IDs↓	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	FM↓
CKDNet-SMTNet [40]	ISPRS2021	DBT	25.0%	47.9%	516	59.5%	40.1%	175	238	16450	42105	1375
TGraM [10]	TGRS2022	JDT	25.1%	43.9%	166	71.7%	31.7%	124	311	7436	51431	1449
DSFNet-SORT [64]	TGRS2023	DBT	27.0%	47.5%	36	83.3%	33.2%	135	352	5084	52413	1045
FairMOT [24]	IJCV2021	JDT	34.7%	55.5%	258	77.4%	43.3%	187	262	8212	42934	2487
TrackFormer [61]	CVPR2022	JDT	34.9%	58.9%	1360	62.7%	55.6%	286	173	20515	29415	1439
TraDeS [25]	CVPR2021	JDT	35.0%	58.6%	118	75.7%	47.9%	202	261	11103	40017	1265
UMA [20]	CVPR2020	JDT	39.8%	62.9%	250	71.6%	56.2%	277	204	15104	32034	1193
StrongSORT [18]	TMM2023	DBT	41.0%	58.8%	**19**	**95.9%**	42.4%	206	312	**1260**	45232	**311**
SGT [62]	WACV2023	JDT	44.8%	61.5%	717	87.7%	47.4%	217	238	3311	39460	1718
SORT [14]	ICIP2016	DBT	47.5%	64.5%	80	92.0%	49.7%	236	254	2525	38726	434
GHOST [63]	CVPR2023	DBT	47.8%	64.9%	162	90.8%	50.5%	239	246	2978	37939	927
CenterTrack [22]	ECCV2020	JDT	54.4%	60.1%	2461	70.3%	52.4%	297	143	6720	26751	640
The Proposed	-	JDT	**61.5%**	**74.3%**	74	87.9%	**64.3%**	**313**	**135**	4583	**25683**	573

method outperforms its baseline on the object localization performance. Moreover, the tracking efficiency is equivalent to its baseline. However, since the ships are densely distributed, the joint training of detection and association may influence the detection performance. Thus, the proposed method cannot outperform the teacher network on AIR-ship. From the training time and memory, even though the KD is introduced, the training cost is not significantly increased, which guarantees the practical application of the proposed method.

E. Experiments on λ_{kd}

λ_{kd} is the parameter we introduce in the objective function to control the weight of the KD module. In order to select the most suitable λ_{kd}. We conduct experiments on both two datasets, namely, AIR-aircraft and AIR-ship. λ_{kd} is selected from 0.1 to 1 with an interval of 0.1. Figs. 6 and 7 show line charts on the main indicators MOTA, IDF1, FP, and FN. Fig. 6 shows the results on the AIR-aircraft dataset. Fig. 7 shows the results on the AIR-ship dataset. As can be seen from Figs. 6 and 7, the overall result showed a trend of first rising and then declining. When λ_{kd} takes a value of 0.8, the best result is obtained on all indicators. Therefore, for all experiments, we select λ_{kd} as 0.8. This also demonstrates that the contribution of KD is obvious on the JDT-based MOT task. Also, the effectiveness of the proposed approach is further demonstrated.

F. State-of-the-Art Comparison

We compare the proposed method with many representative MOT methods, including DBT-based ByteTrack [15], StrongSORT [18], SORT [14], DeepSORT [16], IOU-Tracker [19], SGT [62], GHOST [63], JDT-based UMA [20], CenterTrack [22], and TraDeS [25] on the three datasets, AIR-MOT [41], VISO [6], and SAT-MTB [59]. There are also several methods designed for satellite videos selected for comparison, including CKDNet [40], DSFNet [64], and TGraM [10]. For all DBT-based methods, we use YOLOX-based detectors and randomly select 10% as the training data and the rest as the test data to obtain the detection result. For a fair comparison, the parameters of each compared method follow the default settings in the corresponding papers. Tables V–VII show the quantitative comparisons. Table V shows the results on dataset AIR-MOT. Table VI shows the results on dataset SAT-MTB. Table VII shows the results on dataset VISO.

From Table V, for AIR-aircraft, we can see that the proposed method achieves the best results on MOTA, IDF1, IDR, and MT among all compared methods. In the case of the core indicator MOTA, the proposed method is 5.5% higher than the second place method TGraM [10]. For another important indicator IDF1, the proposed method is 3.2% higher than the second place method TGraM [10]. This demonstrates the state-of-the-art performance of the proposed method on the AIR-MOT dataset on aircraft. For AIR-ship, it can be

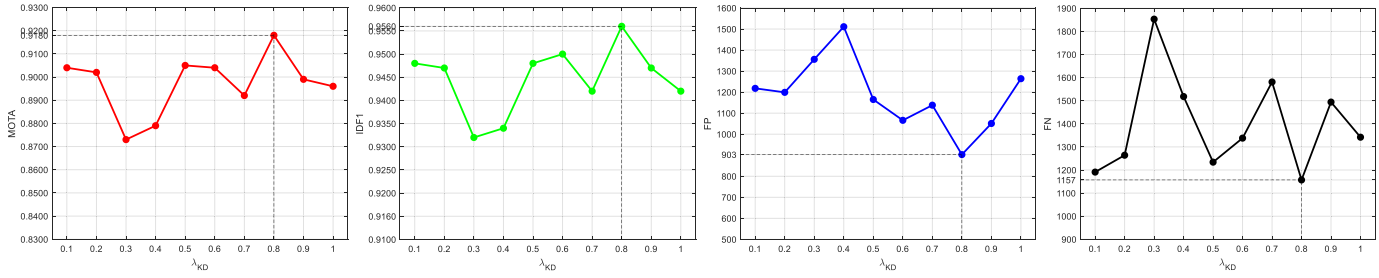
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TABLE VI
COMPARISON WITH STATE-OF-THE-ART APPROACHES ON DATASET SAT-MTB

On **SAT-MTB-ship**.												
Methods	Publication year	DBT/JDT	MOTA↑	IDF1↑	IDs↓	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	FM↓
FairMOT [24]	IJCV2021	JDT	8.2%	45.0%	143	-	-	23	39	6961	12860	-
ByteTrack [15]	ECCV2022	DBT	10.5%	46.9%	168	-	-	38	34	8054	11233	-
DeepSORT [16]	ICIP2017	DBT	13.5%	45.5%	144	-	-	35	40	6012	12641	-
TGraM [10]	TGRS2022	JDT	-13.9%	35.6%	283	-	-	19	47	10103	14383	-
Trackor++ [65]	ICCV2019	DBT	34.3%	56.6%	119	-	-	35	30	3466	10695	-
CKDNet-SMTNet [40]	ISPRS2021	DBT	36.9%	49.8%	299	-	-	43	14	5403	8025	-
DSFNet [64]	TGRS2023	DBT	37.2%	57.7%	235	-	-	38	16	4306	9116	-
CenterTrack [22]	ECCV2020	JDT	54.1%	68.3%	84	69.4%	**68.5%**	39	13	**1864**	5167	325
The Proposed	-	JDT	**57.3%**	**70.5%**	**83**	**71.2%**	68.2%	**43**	**11**	2076	**4023**	**313**
On **SAT-MTB-airplane**.												
Methods	Publication year	DBT/JDT	MOTA↑	IDF1↑	IDs↓	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	FM↓
Trackor++ [65]	ICCV2019	DBT	17.6%	57.8%	64	-	-	33	13	6957	5856	-
FairMOT [24]	IJCV2021	JDT	32.0%	63.9%	1	-	-	32	22	4413	6218	-
DeepSORT [16]	ICIP2017	DBT	35.6%	55.3%	**0**	-	-	22	33	**661**	9400	-
TGraM [10]	TGRS2022	JDT	37.4%	67.5%	6	-	-	33	18	4368	5405	-
ByteTrack [15]	ECCV2022	DBT	45.3%	65.1%	1	-	-	31	26	916	7636	-
CKDNet-SMTNet [40]	ISPRS2021	DBT	54.3%	65.9%	91	-	-	53	**3**	4349	**2700**	-
CenterTrack [22]	ECCV2020	JDT	56.7%	76.9%	38	77.7%	**76.2%**	45	8	3206	3511	**38**
The Proposed	-	JDT	**64.6%**	**77.9%**	66	**81.6%**	74.6%	**48**	8	2053	3404	83

TABLE VII
COMPARISON WITH STATE-OF-THE-ART APPROACHES ON DATASET VISO

Methods	Publication year	DBT/JDT	MOTA↑	IDF1↑	IDs↓	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	FM↓
MMB+FairMOT [66]	TGRS2023	DBT	2.3%	-	**52**	-	-	21	623	2073	83258	205
DSFNet [64]	TGRS2023	DBT	3.6%	-	348	-	-	45	432	24651	86424	1326
MMB+CMOT [66]	TGRS2023	DBT	22.8%	-	89	-	-	38	494	**1**	71638	**111**
CKDNet-SMTNet [40]	ISPRS2021	DBT	33.5%	33.2%	3071	62.8%	22.6%	105	443	**1**	79194	2887
DSFNet+SORT [64]	TGRS2023	DBT	41.6%	-	434	-	-	318	105	32357	37230	8075
MMB+DTTP [66]	TGRS2023	DBT	44.5%	-	3090	-	-	**483**	**22**	38329	**10032**	1344
CenterTrack [22]	ECCV2020	JDT	45.3%	45.7%	5503	**67.3%**	34.6%	275	328	1037	61114	3086
The Proposed	-	JDT	**49.9%**	**48.2%**	5828	65.7%	**38.0%**	329	293	1949	54120	3498



(a) (b) (c) (d)

Fig. 6. Hyperparameter experiments on dataset AIR-aircraft. (a) On metric MOTA. (b) On metric IDF1. (c) On metric FP. (d) On metric FN.

seen that the proposed method owns the best tracking result in terms of MOTA, IDF1, IDR, MT, ML, and FN. This further demonstrates the state-of-the-art performance and generalization performance of the proposed method on AIR-ship targets. From Table VI, for SAT-MTB-airplane, the proposed method owns the best results on MOTA, IDF1, IDP, and ML. This demonstrates the superior performance of the proposed method on dataset SAT-MTB-airplane. In addition, the proposed method achieves the best performance on MOTA, IDF1, IDP, and MT on SAT-MTB-ship. This further demonstrates the state-of-the-art performance of the proposed method on dataset SAT-MTB. From Table VII, the proposed method poses the best performance in terms of MOTA, IDF1, and IDR. This demonstrates the state-of-the-art performance of the proposed method on dataset VISO. However, with more accurate detection performance, more objects can be detected as shown in Fig. 8 by the proposed method, which may also result in relatively more missed associations. As a consequence, the proposed method cannot outperform the compared methods in terms of IDs.

To make the comparison more intuitive, we also visualize some tracking trajectories of top five methods from several

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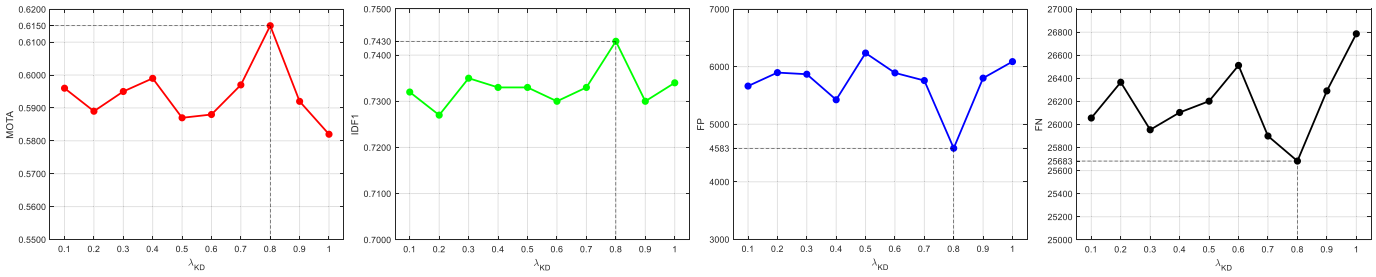


Fig. 7. Hyperparameter experiments on dataset AIR-ship. (a) On metric MOTA. (b) On metric IDF1. (c) On metric FP. (d) On metric FN.

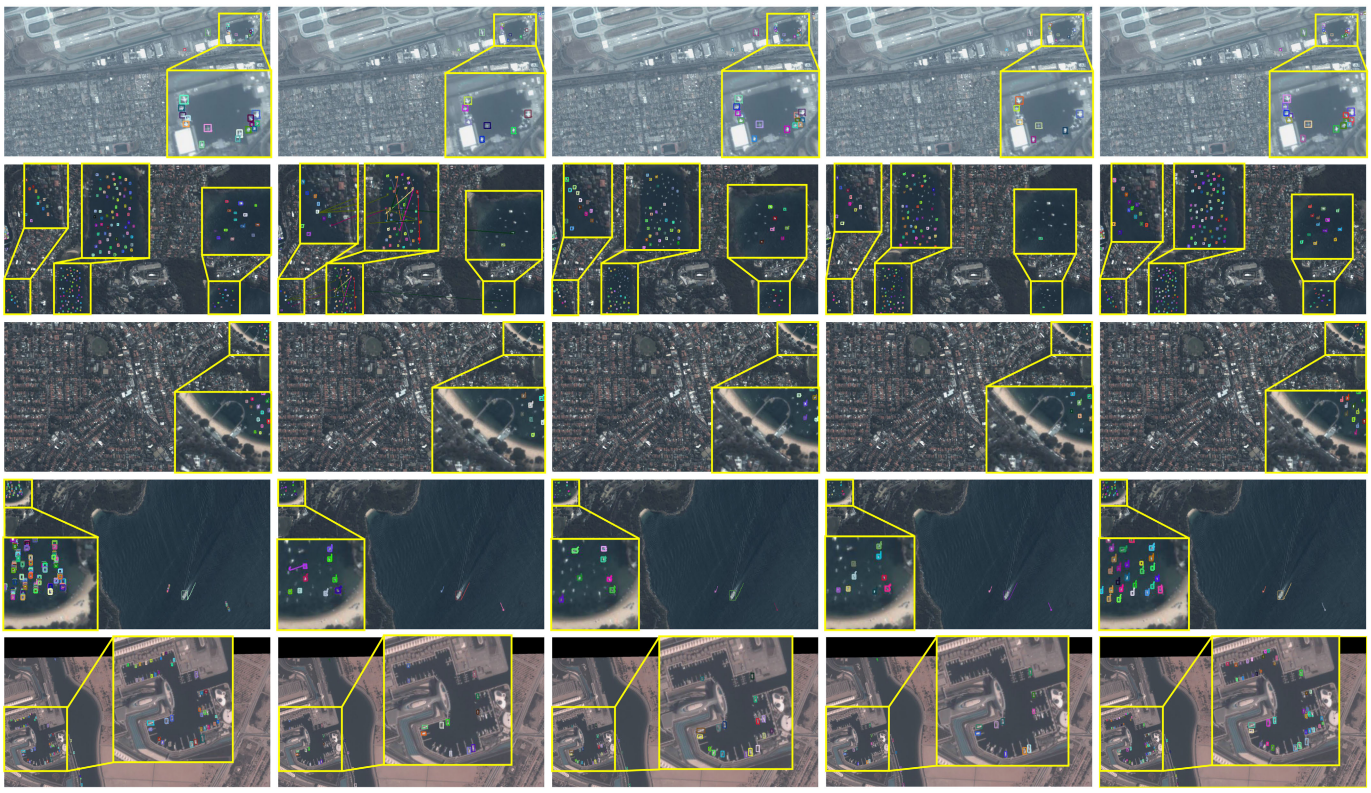


Fig. 8. Selected tracking results visualization of the top five methods, including the proposed method.

videos in Fig. 8. From the visualized results, CenterTrack can detect most objects, but due to the densely distributed objects, the trajectories are not generated and many objects in the next frame are recognized as new objects. GHOST generates many jumping trajectories. There are many mis-detections in the results of SGT and SORT, while for the proposed method, most objects can be detected and it is able to generate continuous trajectories, which demonstrates the superior performance over existing methods.

V. Conclusion and Future Work

In this article, we propose a new tracking framework that inherits both advantages of two- and one-stage tracking frameworks via KD for satellite videos. We design an object distillation method to distill the features of a specialized object detection network based on a one-stage MOT method to improve tracking accuracy while maintaining high tracking speed, which is especially important for satellite videos. Specifically, the adaptive attention-based KD method is designed to distill the features of teacher networks. In addition, in terms of tracklet generation, considering the characteristics of satellite videos, the principle of joint matching based on central point and overlap rate is designed.

From the experimental results, the proposed framework improves the detection performance of it baseline without influencing the tracking efficiency so that state-of-the-art performance is achieved. However, with more detected objects, more missed associations are also introduced. As a consequence, in future work, we will explore SOT-based MOT methods to improve the association's performance.

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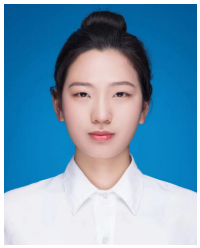
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